

LADYSMITH, South Africa

WMO: 684790

Lat: 28.5756S

Lon: 29.7503E

Elev: 1069

StdP: 89.12

Time Zone: 2.00 (E02)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.1	2.6	-11.6	1.6	16.2	-8.6	2.1	15.3	10.9	14.5	9.1	18.4	1.2	290	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.0	33.8	17.8	32.4	17.8	30.9	17.9	21.6	27.8	20.9	26.8	20.3	26.1	4.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	16.8	23.6	19.2	16.0	23.0	18.6	15.4	22.6	68.6	28.2	65.8	26.6	63.6	26.2	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.3	6.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	22.6	22.7	21.3	18.0	15.6	13.0	12.7	15.2	18.4	19.7	20.9	22.3		
	DBStd	4.80	2.58	2.57	2.69	2.81	2.86	2.73	2.93	3.78	4.45	4.11	3.81	3.32		
	HDD10.0	21	0	0	0	0	1	4	9	4	2	0	0	0		
	HDD18.3	703	3	2	7	38	93	161	175	109	53	35	19	7		
	CDD10.0	3134	389	355	352	241	174	95	93	167	256	301	328	382		
	CDD18.3	772	134	124	101	29	8	1	1	13	56	77	98	131		
	CDH23.3	5866	913	769	618	179	94	19	14	153	574	704	783	1043		
	CDH26.7	2067	308	248	188	30	13	2	0	31	224	286	308	428		
(14) Wind	WSAvg	2.2	2.3	2.0	1.9	1.8	1.3	2.1	1.5	2.1	2.8	2.9	3.0	2.8		
(15) Precipitation	PrecAvg	736	139	106	91	45	14	8	7	24	32	74	90	110		
	PrecMax	1159	309	237	190	225	83	59	75	98	256	179	178	243		
	PrecMin	399	25	14	13	0	0	0	0	0	0	7	5	18		
	PrecStd	190	65	58	46	43	20	13	15	28	47	49	42	50		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	33.9	33.0	29.8	28.5	25.2	24.9	30.3	34.2	35.2	35.3	36.4		
		MCWB	18.8	19.3	19.2	16.0	13.7	11.0	10.6	14.4	15.4	17.2	18.1	18.5		
	2%	DB	32.0	31.7	30.5	27.3	25.9	23.0	23.1	27.1	31.5	32.4	32.5	33.2		
		MCWB	19.2	19.8	18.7	15.7	13.5	10.2	10.2	12.4	14.5	16.1	17.2	18.5		
	5%	DB	30.1	29.7	28.8	25.4	23.8	21.1	21.5	24.9	29.0	30.0	30.2	31.1		
		MCWB	19.3	19.5	18.4	15.6	12.9	10.2	10.1	11.7	13.9	16.0	17.4	18.8		
	10%	DB	28.2	27.9	26.9	23.5	21.6	19.2	19.5	22.7	26.4	27.4	27.9	29.0		
		MCWB	19.3	19.4	18.3	15.2	12.7	10.0	9.5	11.5	13.5	15.8	17.4	18.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	22.4	21.5	18.8	15.9	13.6	13.1	16.3	17.5	20.1	21.1	22.2		
		MCDB	27.9	29.0	28.2	24.9	23.7	20.0	19.7	24.4	27.2	28.5	28.1	28.7		
	2%	WB	21.5	21.5	20.4	17.7	14.9	12.2	12.1	14.8	16.4	18.5	20.0	21.0		
		MCDB	27.1	27.4	26.4	23.1	21.3	19.4	17.9	22.3	25.8	26.3	27.0	27.3		
	5%	WB	20.7	20.8	19.7	17.0	14.1	11.4	11.2	13.6	15.6	17.7	19.1	20.2		
		MCDB	26.3	26.3	25.2	22.0	20.9	17.9	17.8	21.6	24.5	25.3	25.6	26.3		
	10%	WB	20.0	20.2	19.0	16.1	13.3	10.7	10.5	12.5	14.8	17.0	18.4	19.5		
		MCDB	25.6	25.6	24.3	21.0	20.1	16.9	17.3	20.0	23.3	23.9	24.6	25.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	10.7	10.8	11.6	13.6	14.1	14.7	13.8	13.6	12.3	11.9	11.7		
		MCDBR	14.3	13.4	14.0	14.6	16.1	16.3	16.7	16.7	17.8	17.4	16.3	15.5		
	5% WB	MCWBR	4.2	3.8	4.1	4.5	6.2	7.2	7.3	6.3	5.6	4.7	4.1	4.1		
		MCWBR	11.9	11.9	11.8	11.8	14.0	13.4	14.0	14.4	14.8	14.4	13.0	13.0		
(40) Clear-Sky Solar Irradiance	taub	0.391	0.382	0.371	0.335	0.299	0.283	0.288	0.349	0.393	0.427	0.388	0.391			
	taud	2.353	2.382	2.390	2.448	2.512	2.525	2.496	2.332	2.234	2.189	2.329	2.342			
	Ebn at Noon	951	941	920	906	898	894	902	876	883	889	948	954			
	Edh at Noon	133	126	118	101	85	79	85	111	135	151	136	135			
(44) All-Sky Solar Radiation	RadAvg	6.38	6.10	5.38	4.50	3.95	3.58	3.85	4.51	5.30	5.57	6.27	6.53			
	RadStd	0.47	0.51	0.34	0.29	0.20	0.20	0.19	0.27	0.30	0.44	0.43	0.34			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.33	+0.57	N/A	N/A	-0.34	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page